

Date: _____

Day: _____

10-12

Test for Independence

Used to determine whether two categorical variables are independent or related in population.

Formula for expected freq.

For contingency table,

$$E = \frac{(\text{Row Total}) (\text{Column Total})}{\text{Grand Total}}$$

- Row Total: Observed values in row
- Col Total: Observed values in col
- Grand Total: Total no. of obs.

Test Statistic:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

O: Observed Freq

E: Expected Freq

Degree of Freedom:

$$V = (r - 1) (c - 1)$$

r: number of rows

c: number of cols

H_0 : independent

Reject H_0 if

$$\chi^2 > \chi^2_{\alpha(V)}$$

Question:

A political analyst wants to determine whether a voter's opinion on a new tax reform is **independent** of their income level. A random sample of **1,000 registered voters** from Illinois was surveyed, and their responses were categorized based on **income level (Low, Medium, High)** and whether they **support or oppose** the tax reform. The observed data is recorded in the following **contingency table**:

Income Level	For Tax Reform	Against Tax Reform
Low	182	154
Medium	213	138
High	203	110

Using a **Chi-Square Test for Independence**, test whether there is a **significant relationship** between **income level** and **opinion on tax reform** at a **5% significance level** ($\alpha=0.05$).

Solution:			
H_0: Voter opinion on tax reform is independent of income			
H_1: " " " " dependent "			
TABLE			
Income level	For	Against	Total
Low	182	154	336
Medium	213	138	351
Hard	203	110	313
Total	598	402	1000
Expected Frequ			
Table			
Income level	For (Expected)	Against (Expected)	
Low	$\frac{336 \times 598}{1000} = 200.9$	$\frac{336 \times 402}{1000} = 135.1$	
Medium	$\frac{351 \times 598}{1000} = 209.9$	$\frac{351 \times 402}{1000} = 141.1$	
High	$\frac{313 \times 598}{1000} = 187.2$	$\frac{313 \times 402}{1000} = 125.8$	

Chi Squared Statistic:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$$\begin{aligned} \chi^2 = & \frac{(182-200.9)^2}{200.9} + \frac{(213-209.9)^2}{209.9} + \frac{(203-187.2)^2}{187.2} \\ & + \frac{(154-135.1)^2}{135.1} + \frac{(138-141.1)^2}{141.1} + \frac{(110-125.8)^2}{125.8} \end{aligned}$$

$$\chi^2 = 7.85$$

Critical Region.

$$\begin{aligned} v &= (2-1)(3-1) \\ &= 2 \end{aligned}$$

$$\alpha = 0.05 \quad v = 2$$

$$\chi^2_{0.05(2)} = 5.991$$

Since

$$7.85 > 5.991$$

reject H_0 .

Conclusion:

We can't say that voter opinion and income level are independent.